

Lecture 4: GLM Theory — Exponential Family, Moments and Variance Functions

MATH3701 Statistical Modelling

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Course notes: Chapter 3, Sections 3.1–3.5

MATH3701 Statistical Modelling notes, Chapter 3

Why Generalise?

Normal linear models assume:

$$Y_i \sim \mathcal{N}(\mu_i, \sigma^2), \quad \mu_i = \mathbf{x}_i' \boldsymbol{\beta}$$

But many responses are:

- **Counts:** $Y_i \in \{0, 1, 2, \dots\}$ (Poisson)
- **Binary:** $Y_i \in \{0, 1\}$ (Bernoulli)
- **Proportions:** $Y_i/m_i \in [0, 1]$ (Binomial)
- **Positive continuous:** $Y_i > 0$ (Gamma, Exponential)

GLMs: A unified framework for all of these.

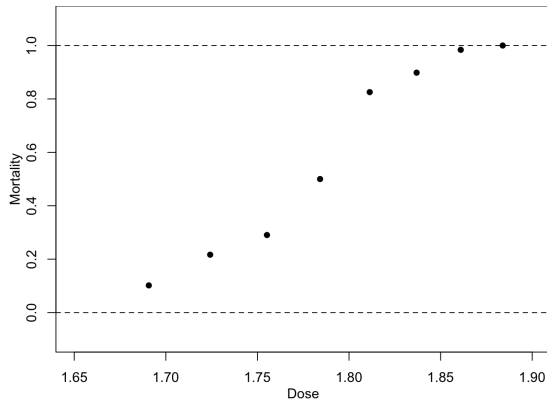
Motivating Example 1: Beetle Mortality

Data: Number killed (Y_i) out of m_i beetles at dose x_i

Model:

$$Y_i \sim \text{Binomial}(m_i, p_i)$$

Goal: Model how p_i depends on dose x_i .



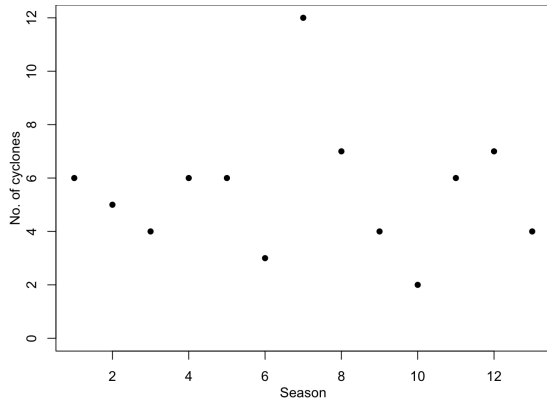
Motivating Example 2: Cyclone Counts

Data: Number of tropical cyclones (Y_i) in season i

Model:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

Goal: Model how λ_i depends on climate variables.



The Three Components of a GLM

A GLM has three parts:

- ① **Random Component:** Distribution of Y from the exponential family
- ② **Systematic Component:** Linear predictor $\eta = \mathbf{X}\beta$
- ③ **Link Function:** Connects mean to linear predictor via $\eta = g(\mu)$

Today: The **random component** (the exponential family and its moments) and the **systematic component**.

Next lecture: The link function.

The Exponential Family

Definition: A distribution belongs to the **exponential family** if its pdf/pmf can be written as:

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

Components:

- θ = **canonical (natural) parameter**
- $\phi > 0$ = **scale (dispersion) parameter**
- $b(\cdot)$ = function determining the distribution
- $c(\cdot, \cdot)$ = normalizing function

Key question: How do we write common distributions in this form?

Poisson Distribution

PMF:

$$f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y = 0, 1, 2, \dots$$

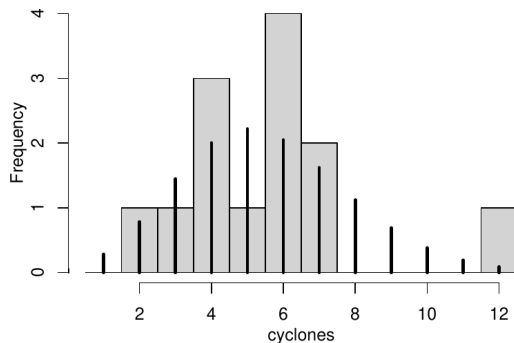
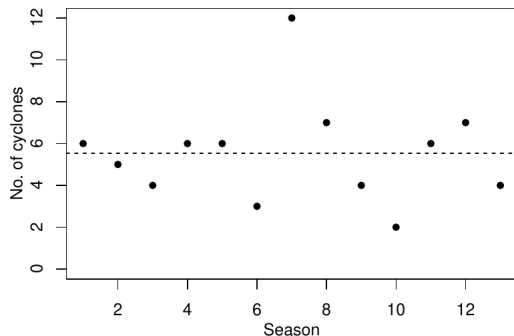
Rewrite using $\lambda^y = e^{y \log \lambda}$:

$$f(y) = \exp\{y \log \lambda - \lambda - \log y!\}$$

Matching to exponential family form:

- $\theta = \log \lambda$ (canonical parameter)
- $b(\theta) = e^\theta = \lambda$
- $\phi = 1$ (no dispersion parameter)
- $c(y, \phi) = -\log y!$

Poisson Fit to the Cyclone Data



Assuming a constant rate $Y_i \sim \text{Poisson}(\lambda)$ across seasons, $\hat{\lambda} = \bar{y}$. The fitted Poisson probabilities (right, bars) match the observed counts (histogram) reasonably well.

Binomial Distribution

PMF:

$$f(y; m, p) = \binom{m}{y} p^y (1-p)^{m-y}, \quad y = 0, 1, \dots, m$$

Rewrite:

$$\begin{aligned} f(y) &= \exp \left\{ y \log p + (m-y) \log(1-p) + \log \binom{m}{y} \right\} \\ &= \exp \left\{ y \log \frac{p}{1-p} + m \log(1-p) + \log \binom{m}{y} \right\} \end{aligned}$$

The key step: factor out to get y times a function of p .

Binomial: Exponential Family Components

From the rewritten form:

$$f(y) = \exp \left\{ y \log \frac{p}{1-p} + m \log(1-p) + \log \binom{m}{y} \right\}$$

Matching to exponential family:

- $\theta = \log \frac{p}{1-p} = \text{logit}(p)$ (canonical parameter)
- $b(\theta) = m \log(1 + e^\theta)$
- $\phi = 1$
- $c(y, \phi) = \log \binom{m}{y}$

Useful identity:

$$p = \frac{e^\theta}{1 + e^\theta}, \quad 1 - p = \frac{1}{1 + e^\theta}$$

Normal Distribution

PDF:

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}$$

Expand and rewrite:

$$\begin{aligned} f(y) &= \exp \left\{ -\frac{y^2 - 2y\mu + \mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\} \\ &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\} \end{aligned}$$

Group terms: $y\theta - b(\theta)$ in the numerator, rest in $c(y, \phi)$.

Normal: Exponential Family Components

Matching to exponential family form:

- $\theta = \mu$ (the canonical parameter equals the mean)
- $b(\theta) = \theta^2/2 = \mu^2/2$
- $\phi = \sigma^2$ (dispersion parameter)
- $c(y, \phi) = -\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi)$

Special property of Normal:

- The canonical parameter θ is just μ
- This is why normal linear models use $\mu = \mathbf{x}'\boldsymbol{\beta}$ directly

Summary: Exponential Family Forms

Distribution	θ	$b(\theta)$	ϕ	$c(y, \phi)$
Poisson(λ)	$\log \lambda$	e^θ	1	$-\log y!$
Binomial(m, p)	$\text{logit}(p)$	$m \log(1 + e^\theta)$	1	$\log \binom{m}{y}$
Normal(μ, σ^2)	μ	$\theta^2/2$	σ^2	$-\frac{y^2}{2\phi} - \frac{1}{2} \log(2\pi\phi)$

Key observation: Different distributions share the same unified form.

Why the Exponential Family?

Advantages of the exponential family form:

- ① **Unified theory:** Same formulas work for all members
- ② **Moments from derivatives:** Mean and variance can be found from $b(\theta)$
- ③ **Maximum likelihood:** Score equations have nice form
- ④ **Sufficient statistics:** Data can be summarized efficiently

Members: Normal, Poisson, Binomial, Gamma, Exponential, Geometric, Inverse Gaussian, ...

Not members: Student's t , Uniform (with unknown bounds)

Summary: The Exponential Family

Key points:

- GLMs extend linear models to non-normal responses
- GLMs have three components: random, systematic, link
- The random component uses the exponential family
- Exponential family form: $f(y) = \exp\{(y\theta - b(\theta))/\phi + c(y, \phi)\}$
- Poisson: $\theta = \log \lambda$, $b(\theta) = e^\theta$
- Binomial: $\theta = \text{logit}(p)$, $b(\theta) = m \log(1 + e^\theta)$
- Normal: $\theta = \mu$, $b(\theta) = \theta^2/2$

Moments of Exponential Family Distributions

Proposition

For a distribution in exponential family form:

$$\mathbb{E}[Y] = b'(\theta), \quad \text{Var}[Y] = b''(\theta) \cdot \phi$$

Key insight: The mean and variance are determined entirely by derivatives of $b(\theta)$.

This is why the exponential family is so powerful — once we know $b(\theta)$, we know the moments.

Proof Sketch: Deriving the Mean

Starting point: All probability densities integrate to 1:

$$\int f(y; \theta, \phi) dy = 1$$

Differentiate both sides w.r.t. θ :

$$\frac{\partial}{\partial \theta} \int f(y) dy = 0$$

Move derivative inside:

$$\int \frac{\partial f}{\partial \theta} dy = \int f(y) \cdot \frac{y - b'(\theta)}{\phi} dy = 0$$

Rearranging:

$$\mathbb{E}[Y] = b'(\theta)$$

Proof Sketch: Deriving the Variance

Differentiate again w.r.t. θ :

Taking the second derivative of $\int f(y) dy = 1$ and simplifying gives:

$$\mathbb{E} \left[\left(\frac{Y - b'(\theta)}{\phi} \right)^2 \right] = \frac{b''(\theta)}{\phi}$$

Since $\mathbb{E}[Y] = b'(\theta)$:

$$\frac{\text{Var}[Y]}{\phi^2} = \frac{b''(\theta)}{\phi}$$

Therefore:

$$\text{Var}[Y] = b''(\theta) \cdot \phi$$

Verifying Moments: Poisson

Recall: $\theta = \log \lambda$, $b(\theta) = e^\theta$, $\phi = 1$

Mean:

$$\mathbb{E}[Y] = b'(\theta) = \frac{d}{d\theta} e^\theta = e^\theta = \lambda \quad \checkmark$$

Variance:

$$\text{Var}[Y] = b''(\theta) \cdot \phi = e^\theta \cdot 1 = \lambda \quad \checkmark$$

Note: For Poisson, $\mathbb{E}[Y] = \text{Var}[Y] = \lambda$ (equidispersion).

Verifying Moments: Binomial

Recall: $\theta = \text{logit}(p)$, $b(\theta) = m \log(1 + e^\theta)$, $\phi = 1$

Mean:

$$b'(\theta) = m \cdot \frac{e^\theta}{1 + e^\theta} = mp \quad \checkmark$$

Variance:

$$\begin{aligned} b''(\theta) &= m \cdot \frac{e^\theta \cdot (1 + e^\theta) - e^\theta \cdot e^\theta}{(1 + e^\theta)^2} = m \cdot \frac{e^\theta}{(1 + e^\theta)^2} \\ &= mp(1 - p) \quad \checkmark \end{aligned}$$

Note: Variance depends on mean — this is the *mean-variance relationship*.

Verifying Moments: Normal

Recall: $\theta = \mu$, $b(\theta) = \theta^2/2$, $\phi = \sigma^2$

Mean:

$$\mathbb{E}[Y] = b'(\theta) = \frac{d}{d\theta} \frac{\theta^2}{2} = \theta = \mu \quad \checkmark$$

Variance:

$$\text{Var}[Y] = b''(\theta) \cdot \phi = 1 \cdot \sigma^2 = \sigma^2 \quad \checkmark$$

Note: For Normal, variance is independent of mean.

The Variance Function

Define: The **variance function** $V(\mu)$ by:

$$\text{Var}[Y] = V(\mu) \cdot \phi$$

Distribution	Mean μ	Variance function $V(\mu)$
Poisson	λ	μ
Binomial	mp	$\mu(1 - \mu/m)$
Normal	μ	1
Gamma	α/λ	μ^2

The variance function characterizes how spread changes with the mean.

Summary: Exponential Family Moments

Distribution	θ	$b(\theta)$	ϕ	$\mathbb{E}[Y]$	$\text{Var}[Y]$
Poisson	$\log \lambda$	e^θ	1	λ	λ
Binomial	$\text{logit}(p)$	$m \log(1 + e^\theta)$	1	mp	$mp(1 - p)$
Normal	μ	$\theta^2/2$	σ^2	μ	σ^2

Key formula: $\mathbb{E}[Y] = b'(\theta)$, $\text{Var}[Y] = b''(\theta)\phi$

The Systematic Component

Now we connect the distribution to explanatory variables.

Linear predictor:

$$\eta_i = \sum_{j=1}^p \beta_j x_{ij} = \mathbf{x}_i' \boldsymbol{\beta}$$

In matrix form:

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}$$

where:

- $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)'$ is the $n \times 1$ vector of linear predictors
- $\mathbf{X}$ is the $n \times p$ design matrix
- $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)'$ is the parameter vector

Normal Linear Model as a GLM

Random component: $Y_i \sim \mathcal{N}(\mu_i, \sigma^2)$

Systematic component: $\eta_i = \mathbf{x}_i' \boldsymbol{\beta}$

Link function: Identity link, $g(\mu) = \mu$

So:

$$\mu_i = \eta_i = \mathbf{x}_i' \boldsymbol{\beta}$$

This is exactly the normal linear model we studied earlier.

$\Rightarrow$ Normal regression is a *special case* of GLM.

Summary: Moments and Model Structure

Key points:

- Moments from $b(\theta)$: $\mathbb{E}[Y] = b'(\theta)$, $\text{Var}[Y] = b''(\theta)\phi$
- Variance function $V(\mu)$ describes mean-variance relationship
- Systematic component: linear predictor $\eta = \mathbf{X}\boldsymbol{\beta}$
- Normal regression is a GLM with identity link

Next lecture: The link function — canonical links, then logit, probit and complementary log-log.